

21. Answer: A

Power dissipated $P = I^2 R$

$$400 \times 10^{-3} = (40 \times 10^{-3})^2 \times R$$

$$R = 250 \, \Omega$$

Using $R = \frac{\rho L}{A}$

$$250 = \frac{\rho \times 1.8 \times 10^{-2}}{1.5 \times 10^{-2} \times 1.2 \times 10^{-3}}$$

$$\rho = 0.25 \, \Omega \text{ m}$$